MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG1MDCECT103 - Audio Electronics

Duration: 1 hour **Maximum Marks: 35**

Answer *any 35* questions

Each Question Carry ONE Mark

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

** Understand (U), Apply (A), Analyse (An), Create (C), Skill (S)*

1	What property of sound is measured in decibels (dB) A) Wavelength B) Amplitude C) Frequency D) Velocity	U	1
2	What is the term for the quality of a sound that distinguishes it from other sounds of the same pitch and volume? A) Dynamics B) Timbre C) Harmony D) Resonance	U	1
3	What phenomenon occurs when sound waves reflect off a surface? A) Echo B) Absorption C) Diffraction D) Interference	U	1
4	Which property of sound determines its pitch? A) Amplitude B) Frequency C) Wavelength D) Velocity	U	1
5	Which of the following factors can affect the speed of sound in a medium? A) Temperature B) Density of the medium C) Humidity D) All of the above	U	1
6	Which of the following describes the frequency of a sound wave? A) The loudness of the sound B) The speed of sound C) The number of cycles per second D) The duration of the sound	U	1
7	What is an interval in music? A) The duration of a note B) The distance between two pitches C) The speed of sound D) The loudness of a note	U	1
8	Which term describes a sound wave that consists of a fundamental frequency and its multiples? A) Interval B) Harmonics C) Timbre D) Phase	U	1
9	How many semitones are there in an octave? A) 5 B) 7 C) 12 D) 14	U	1
10	If the frequency of a sound is doubled, what happens to its pitch? A) It remains the same B) It decreases C) It increases D) It becomes inaudible	U	1
11	What is the primary function of a preamplifier in an audio system? A) To amplify low-level audio signals to a usable level B) To convert audio signals into digital format C) To mix multiple audio signals into one output D) To provide power to the speakers	U	2
12	Which of the following statements about power amplifiers is true? A) They are used primarily for signal processing B) They can drive speakers directly to produce sound C) They only work with digital audio signals. D) They have a lower power output than preamplifiers	U	2

13 In an audio system, where is the preamplifier typically located? U 2
A) After the power amplifier B) Before the signal reaches the power amplifier
C) Inside the speakers D) In the recording device only

14 What is a common characteristic of a power amplifier? U 2
A) It typically has a low gain compared to a preamplifier.
B) It operates at higher voltages and currents.
C) It is designed to modify audio signals significantly.
D) It is always built into the audio source device.

15 What type of signals do preamplifiers usually handle? U 2
A) High-level speaker signals.
B) Low-level audio signals from sources like microphones and instruments.
C) Digital audio signals only.
D) Radio frequency signals.

16 What is a digital amplifier primarily used for? U 2
A) To amplify audio signals using analog components.
B) To convert digital audio signals into analog for playback.
C) To directly amplify digital signals without converting them to analog.
D) To process video signals in television systems.

17 What is the term used to describe the process by which a digital amplifier adjusts its output based on the incoming signal? U 2
A) Modulation B) Sampling C) Feedback D) Encoding

18 Which of the following describes a key advantage of digital amplifiers compared to traditional analog amplifiers? U 2
A) Higher power consumption. B) Increased distortion at high volumes.
C) Greater efficiency and reduced heat generation. D) More complex circuitry.

19 Which type of digital amplifier is known for high efficiency and low distortion? U 2
A) Class A amplifier B) Class B amplifier
C) Class D amplifier D) Class AB amplifier

20 Digital amplifiers often use which technology to convert signals? U 2
A) Class A amplification B) Pulse Width Modulation (PWM)
C) Linear voltage regulation D) Frequency modulation

21 Which connector is often used for professional audio applications and provides a balanced signal to reduce noise? U 2
A) HDMI B) SPDIF C) XLR D) RCA

22 MIDI is primarily used for what type of communication? U 2
A) Transmitting video signals B) Connecting microphones to mixers
C) Sending digital music performance data D) Streaming high-definition audio

23 What is the primary purpose of the SPDIF connector? U 2
A) To connect speakers to an amplifier.
B) To transmit digital audio signals over short distances.
C) To interface musical instruments with computers.
D) To provide video and audio signals simultaneously.

24 Which of the following connectors is capable of carrying both audio and video signals? U 2
A) TRS B) HDMI C) XLR D) RCA

25 What does TRS stand for in the context of audio connectors? U 2
A) Twin Ring Socket B) Tip-Ring-Sleeve
C) Transistor-Resistor-Speaker D) Tone-Resonance-Signal

26 What is the main advantage of using balanced cables over unbalanced cables? U 2
A) They are cheaper to manufacture.
B) They have a higher impedance.
C) They reduce noise and interference over long distances.
D) They are easier to connect.

27	Which of the following connectors is typically associated with balanced wiring? A) RCA B) TRS C) XLR D) Both B and C	U	2
28	In an unbalanced cable, which of the following typically carries the audio signal? A) Positive wire only B) Positive and negative wires C) Ground wire only D) Shield wire only	U	2
29	In a typical audio mixer block diagram, which component is responsible for adjusting the volume of each channel? A) Equalizer B) Fader C) Bus D) Master Output	U	2
30	What is the purpose of a bus in a mixer? A) To amplify individual channels. B) To combine multiple audio signals into one output. C) To provide a visual representation of sound levels. D) To isolate channels from one another.	U	2
31	At which frequency range is the human ear most sensitive? A) 20-500 Hz B) 500 Hz - 4 kHz C) 10 kHz - 20 kHz D) 100 Hz - 10 kHz	U	1
32	What is the main function of the ossicles in the middle ear? A) To convert sound into electrical signals B) To amplify and transmit sound from the outer ear to the inner ear C) To filter out low-frequency noise D) To protect the cochlea from loud sounds	U	1
33	Which phenomenon describes the ear's reduced sensitivity to low- and high-frequency sounds compared to mid-frequency sounds at low volumes? A) Acoustic masking B) Equal loudness contour C) Phase cancellation D) Doppler effect	U	1
34	Which term is used to describe a condition where specific frequencies sound less intense due to interference from other frequencies? A) Loudness adaptation B) Acoustic reflex C) Auditory masking D) Frequency tuning	U	1
35	In audiometry, the 'threshold of hearing' refers to: A) The quietest sound the human ear can detect at a specific frequency B) The loudest sound before pain occurs C) The range between the softest and loudest sounds detected D) The range of frequencies detected by the ear	U	1
36	What is the typical role of the auditory nerve in the human ear? A) To filter out background noise B) To convert mechanical vibrations into electrical signals C) To carry electrical signals from the cochlea to the brain D) To amplify sound	U	1
37	Which of the following is a common use for digital amplifiers? A) Home theatre systems B) Medical imaging equipment C) Oscilloscopes D) Signal generators	U	2
38	In a digital amplifier, which component is primarily responsible for converting the digital signal to an amplified audio output? A) Digital-to-Analog Converter (DAC) B) Modulator C) Output stage D) Transistor array	U	2
39	Which of the following is generally a limitation of digital amplifiers compared to analog amplifiers? A) Reduced efficiency at lower power outputs B) Limited audio fidelity at very high frequencies C) Increased heat generation D) Larger physical size	U	2
40	Digital amplifiers are often categorized under which class of amplifiers? A) Class A B) Class B C) Class D D) Class AB	U	2

1 * 35 = 35